Nadira Farjana

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Research Interest

Computational Materials Science | Density Functional Theory (DFT) | Lead-Free Halide Materials | Machine learning and AI-assisted materials modeling | Optoelectronic & Photovoltaic Materials | Fabrication of Perovskite Solar Cell

Education

BSc. in Electrical and Electronic Engineering | Jamalpur Science & Technology University | CGPA: 3.63/4.00 (Class Rank: 3rd out of 33 students)

Higher Secondary Certificate (H.S.C) | Shahid Syed Nazrul Islam College, Mymensingh | GPA: 5.0/5.0 | Year: 2020

Secondary School Certificate (S.S.C) | Muslim Girls' High School & College, Mymensingh |GPA: 5.0/5.0 | Year: 2018

Research Experience

Undergraduate Researcher (2024 – Present)

- Conducting first-principles Density Functional Theory (DFT) simulations using Quantum ESPRESSO to investigate structural, electronic, optical, mechanical, thermal, and phonon properties of semiconductor materials.
- Performing computational simulations using SCAPS-1D to investigate material and solar cell properties.
- Analyzing simulation results using Python, xmgrace and OriginPro.

Publications and Manuscript

Published Journal Article

- Khan, R., **Farjana, N.,** Jim, M. J. A., Al-Humaidi, J. Y., Islam, M. R., & Rana, M. M. (2025). Numerical simulation and performance enhancement of $CsBi_3I_{10}$-based heterojunction solar cell with various semiconductor layers (CZTS, CZTGS, $Al_{0.8}Ga_{0.2}Sb$, GaAs) along with machine learning-based analysis. *Solar Energy*, 295, 113539. DOI: 10.1016/j.solener.2025.113539

Manuscripts Submitted

- Farjana, N., et al. " Device modeling and performance analysis of $CsGeI_2Br$-based single junction and $CsGeI_2Br$/GaAs-based heterojunction perovskite solar cell with appropriate charge transport layers." (Journal Article - Luminescence, Co-Author)
- Farjana, N., et al. " Computational Analysis, Performance Enhancement and Comparison of Cs_2SnBr_6, $ZnSnN_2$, $Cs_2SnBr_6/ZnSnN_2$-based Perovskite Solar Cells " (iCONNECT 2026, 1st Author)
- Farjana, N., et al. " An Integrated Assistive System for Visually Impaired Individuals Using Autonomous Navigation, Deep Learning Currency Recognition, and ReaMe l-Time GPS Tracking " (WIECONECE 2026, 1st Author)

Manuscript under preparation (all first author)

- Farjana, N., et al. " First–Principles Investigation of Composition-Driven Structural, Electronic, Optical, Mechanical, and Thermal Properties in Mixed-Halide $Mg3PCl_{3-x}Br_x$ (where x = 0, 1, 2 and 3) Materials." (Thesis Project, 80% Completed for submission)
- Journal manuscript, based on SCAPS-1D and machine learning, in Simulation stage (50% Simulation done).
- Journal manuscript, based on Bangladeshi banknote detection with deep learning (Completed data collection, now at Leveling stage).

Poster Presentation

- Poster Presentation: Smart Cow Health Monitoring Using IoT for Urban Dairy Farming, presented at the 1st International Conference on Agricultural Machinery and Bioresource Engineering (ICAMBE 2025), Bangladesh Agricultural University, Mymensingh, Bangladesh, 12–13 February 2025.

Awards & Honors

- Third Position – Firm Mechanization Idea Competition, Site event of Seminar on Farm Machinery (2025)
- Second Position – Agricultural Machineries exhibition and talent exploration idea contest (2024)
- Champion – Project Showcasing, JSTU EEE Day (2024)
- First Position – Innovation Showcasing Program, E-Governance and Innovation Committee, JSTU (2024)
- Champion – Project Showcasing, National Conference on Robotics and IoT, Dept. of CSE, JSTU (2023)
- Champion – Project Showcasing, JSTU EEE Day (2023)

Capstone Projects

- Self-Driving Car with Automatic Currency Counter and Barcode Reader for the Blind Individuals (April 2023 – Present)
- Design, Set up and Analysis of Off-Grid Solar System (May 13–26, 2024)

Notable Engineering Projects

GPS Vehicle Tracking System (Arduino, GPS, GSM) | Environment Monitoring System (ESP32, MQ-4, Temperature Sensor, Bluetooth) | Solar Tracker (Arduino, LDR, Servo Motor) | Protection System from Cylinder Disaster (Arduino, Automation, Servo Motor, GSM) | Smart Agricultural System (IoT, Solar, Automation)

Industrial Training & Visits

- Bangladesh Power Development Board (BPDB), Rajshahi Training Centre (Including Chapainawabganj 100 MW HFO Power Plant Visit) | Duration: 7 Days | Year: May 2026
- Jamalpur Palli Bidyut Samity Substation, Beltia, Bangladesh | EEE Departmental Visit, Jamalpur Science & Technology University, August 2025

Technical Skills

- **Technical: Programming:** C++, Python, MATLAB | **Embedded Systems:** Arduino Uno, ESP32, Raspberry Pi | **Simulation & Design:** Quantum ESPRESSO, Vesta, MATLAB/Simulink, Proteus, OrCAD, SCAPS-1D, AutoCAD, OriginPro.
- **Soft Skills:** Leadership & Teamwork, Analytical Problem Solving, Technical Communication.
- **Languages:** Bengali – Native, English – Professional.

Leadership & Extracurricular Activity

- Vice Chair (Activity) in IEEE Student Branch JSTU | Student Member, IEEE Women in Engineering
- Red Crescent Youth (RCY) at Jamalpur Red Crescent Society
- Co-Mentor – RERC (Renewable Energy Research Center), Jamalpur Science & Technology University.
- Executive Member and Coordinator – JSTU Research Society, Jamalpur Science & Technology University.
- Monitoring & Data Analysis Officer, National Taskforce & International Alliance For El Niño & Heatwave Preparedness, Rapid Response BD